

AE-363: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 1 mM KOH

Date: 2024-10-17

Tags: Dihydroxo Radiation PNP
P(tBu)N(py)P(tBu) [Ru(CO)(OH)₂(PNP)]
NMR AE 1H 31P

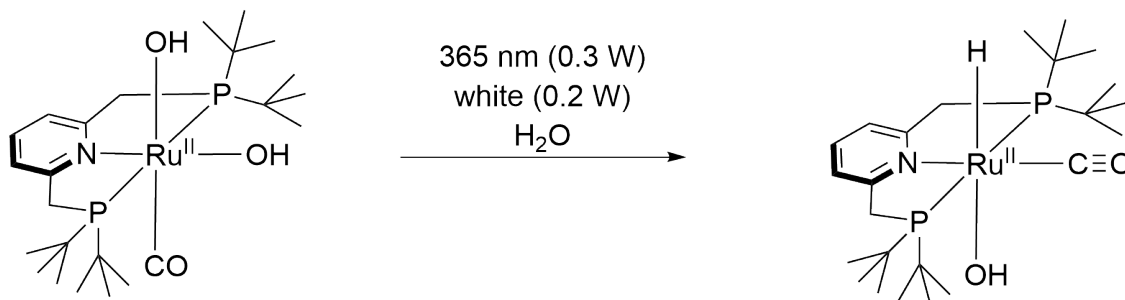
Category: Two-photon

Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: C₂₄H₄₅NO₃P₂Ru

Molecular Weight: 558,64552

[Ru(CO)(OH)₂(PNP^{tBu})]

Chemical Formula: C₂₄H₄₅NO₂P₂Ru

Molecular Weight: 542,64652

[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiment	AE-348: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, water, longer irradiation Two-photon - AE-357: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	AE-310: Synthesis of [Ru(CO)(OH) ₂ (PNP ^{tBu})] using Ag ₂ O, 5 h (AE-310-1)	4.48	/	2.50	2.45	558.65	/
KOH (5 mM)	Prep work - AE-361: 5 mM Stocksolution of KOH in water	/	/	/	/	/	0.18
Milli-Q water	KRA-104: Degassing of Milli-Q water	/	/	/	/	/	0.72

Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9.5
Thermostat temperature [°C]	25

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#). All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
17.10		A 10 mL flask was prepared	
	13:30	To the flask [Ru] was added under air	
		The flask was set under argon	
	14:45	To the flask KOH (5 mM) was added	
	14:55	The obtained mixture was stirred at 320 rpm.	right.jpg
	15:05	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (AE-363-1) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm)	AE-363-1.jpg
21.10		The setup from Two-photon - AE-362: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 5 mM KOH was used without any modification	
	13:33	The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	
	13:34 - 16:35	The J.Young NMR tube was irradiated for 3:01 h	irrad setup.jpg
	16:50	An NMR was measured (AE-363-2)	AE-363-2.jpg

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
18.10	14:26	AE-363-1	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-363-1.zip	AE-363-2_10.nmrium	/	As expected
22.10	20:56	AE-363-2	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-363-2.zip	AE-363-2_10.nmrium	/	Prod seen, as well as main side product, but still mainly substrate

Results

Sample	Yield (hydride species) [%]	Yield (dihydroxo) [%]	Data
AE-363-2	10	43	AE-363-2_Yield_Determination.xlsx

For comaprison with other experiments see: [Two-photon - AE-371: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 0.1 mM KOH](#)

Linked experiments

- [AEI-022: Radiation of \[Ru\(CO\)\(OH\)2\(PNN\)\], NO2-route, 365 nm and white LED](#)
- [AEI-026: Capillary production](#)
- [AEI-064: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)
- [AEI-067: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], crude, 365 nm and white](#)
- [AEI-071: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purer, 365 nm and white](#)
- [AE-105: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)
- [AE-110: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 365 nm and white](#)
- [AE-119: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 52 mM KOH, 365 nm and white](#)
- [AE-140: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 365 nm and white, longer duration I](#)
- [AE-145: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)
- [AE-158: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purified, 365 nm and white](#)

- KRA-092: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$, 365 nm and white

- KRA-104: Degassing of Milli-Q water

- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-243: 1 M Stocksolution of KOH in water

- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white

- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

Prep work - AE-361: 5 mM Stocksolution of KOH in water

Two-photon - AE-348: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, water, longer irradiation

Two-photon - AE-349: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation

Two-photon - AE-357: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1000 mM KOH

Two-photon - AE-358: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH

Two-photon - AE-362: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 5 mM KOH

Two-photon - AE-371: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 0.1 mM KOH

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

Attached files

AE-363-2_10.nmrium

sha256: a596a502cb3ada82e69c66a53f0838b84da871419cc3b8cfd250b5b7ad6275d9

AE-363-2_Yield_Determination.xlsx

sha256: 871345e820c947a7be91062c056fa0c9ee1d8c6c93738e6d4473485a87f2fe02

AE-363-2.zip

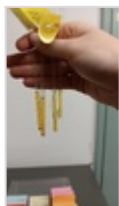
sha256: 8bf4ab8907b2f9976121a71f5dd9a72dc7f8dec9eb91f70ec608ab65c1862747

AE-363-1.zip

sha256: e68595c7ac143304952da64968e595e31c1c6719366d22bc2de83a044b77a5d5

comparison-after-irrad.jpg

sha256: 1efd569a891010ed22145bb4cb9276e25b6930bf8974dafa6e9bba5dc5e5ab99



AE-363-2.jpg

sha256: 5886eba73806761f9b60b04f4ac8490ca0a2c19b38527ccb8503d0a0f3e80247



irrad-setup.jpg

sha256: 8437382b512f67305fdbc377831c8e81ab0349fa59a5e0297ff3c1c25b9a4bf1



AEI-06711.cdxml

sha256: 65d02cd9f4a21042d9bbb3b568aacc15b4f98db9ed95de74a4e781b0391592

AE-363-1.jpg

sha256: 68cff25e195b9f6627c6ccf82cb6f428850806193f31efe71d5a427118485e37



right.jpg

sha256: 1f767bce0a24d83e3f9c21376dc18e1fec3a587507c2285f3e023449a33058e4



Unique eLabID: 20241018-31e1a35fb08405617ca42d5042c2fd4931ff3ec8

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1432>